

Figure 1: Hard drive costs per GB of data (Source: <http://www.mk.com/cost-per-gigabyte>)

corporates and enterprise level customers are generating huge volumes of data, and to have backups is critical for them.

Backing up data is very important. After taking a backup, we have to also make sure that this data is secure, is manageable and that the data's integrity is not compromised. Keeping in mind these aspects, many open source backup software have been developed over a period of years.

Data backup comes in different flavours like individual files and folders, whole drives or partitions, or full system backups. Nowadays, we also have the 'smart' method, which automatically backs up files in commonly used locations (syncing) and we have the option of using cloud storage.

Backups can be scheduled, running as incremental, differential or full backups, as required.

For organisations and large enterprises that are planning on selecting backup software tools and technologies, this article reviews the best open source tools. Before choosing the best software or tool, users should evaluate the features they provide, with reference to stability and open source community support.

Advanced open source storage software like Ceph, Gluster, ZFS and Lustre can be integrated with some of the popular backup tools like Bareos, Bacula, AMANDA and CloneZilla; each of these is described in detail in the following section.

Ceph

Ceph is one of the leading choices in open source software for storage and backup. Ceph provides object storage, block storage and file system storage features. It is very popular because of its CRUSH algorithm, which liberates storage clusters from the scalability and performance limitations imposed by centralised data table mapping. Ceph eliminates many tedious tasks for administrators by replicating and rebalancing data within the cluster, and delivers high performance and infinite scalability.

Ceph also has RADOS (reliable autonomic distributed object store), which provides the earlier described object, block and file system storage in singly unified storage clusters. The Ceph RBD backup script in the v0.1.1 release of `ceph_rbd_bck.sh` creates the backup solution for Ceph.

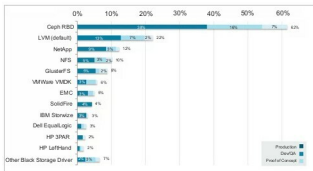


Figure 2: Ceph adoption rate (Source: <https://sanenthusiast.com/top-5-storage-data-center-tech-predictions-2016/>)

This script helps in backing up Ceph pools. It was developed keeping in mind backing up of specified storage pools and not only individual images; it also allows retention of dates and implements a synthetic full backup schedule if needed.

Many organisations are now moving towards large scale object storage and take backups regularly. Ceph is the ultimate solution, as it provides object storage management along with state-of-art backup. It also provides integration into private cloud solutions like OpenStack, which helps one in managing backups of data in the cloud.

The Ceph script can also archive data, remove all the old files and purge all snapshots. This triggers the creation of a new, full and initial snapshot.

OpenStack has a built-in Ceph backup driver, which is an intelligent solution for VM volume backup and maintenance. This helps in taking regular and incremental backups of volumes to maintain consistency of data. Along with Ceph backup, one can use a tool called CloudBerry for versatile control over Ceph based backup and recovery mechanisms.

Ceph also has good support from the community and from large organisations, many of which have adopted it for storage and backup management and intend contribute back to the community.

A lot of developments and enhancements are happening on a continuous basis with Ceph. A number of research organisations have predicted that Ceph's adoption rate will increase in the future. Ceph also has certain cost advantages in comparison with other software products.

More information about the Ceph RBD script can be found at <http://obsidiancreeper.com/2017/04/03/Updated-Ceph-Backup/>.

Gluster

Red Hat's Gluster is another open source software defined scale out, backup and storage solution. It is also called RGHS. It helps in managing unstructured data for physical, virtual and cloud environments. The advantages of Gluster software are its cost effectiveness and highly available storage that does not compromise on scale or performance.